LV VFDs Data Sheet

7043-DSH-COX-950-734-0001
06/07/2026

Revision Control Sheet

Revision	Date	Purpose	Reason for revision	Issued	Reviewed	Approved
0	06/07/2026	Information	First Issue	MPS	FJDM	ESS

Reference Documents	
1	7043-SPE-COX-950-734-0001_LV_VFD_TS
2	7043-SPE-COX-950-702-0001 Electrical motors TS
3	7043-SPE-COX-950-508-0001 Secondary Pumps_TS
4	7043-DSH-COX-270-512-0001 RO CIP&Flushing Pump_DSH Technical Datasheet
5	7043-DSH-COX-260-504-0001 ERD Booster Pumps_DSH Technical Datasheet
6	7043-DSH-COX-230-511-0001 Backwash Pump_DSH Technical Datasheet
7	7043-DSH-COX-230-547-0001 Blowers_DSH Technical Datasheet
8	Annex II: LV VFD Signals
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MODON ORASCOM CONSTRUCTION TROJAN COX

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KKS	Units	Voltage (V)	Type	Rated Power (kW)
TBD	4	400 V	ERD Booster Pumps	160 kW
TBD	2	400 V	Washing Blowers	160 kW
TBD	2	400 V	Backwash Pumps	200 kW
TBD	3	400 V	RO CIP & Flushing Pumps	315 kW

MODON ORASCOM CONSTRUCTION TROJAN COX

LV frecuency converter DSH (ERD Booster pumps)

Item	Description	Requested data	Offer	Offered data (*)
1.	General			
1.1.	Supplier	(*)		Schneider Electric
1.2.	Catalogue No.	(*)		
1.3.	KKS code	TBD		
1.4.	Manufacturer	(*)		Schneider Electric
1.5.	Place of manufacture	(*)		
1.6.	Year of manufacture	(*)		2026 or 2027
1.7.	Quantity	4		
1.8.	Applicable standards	IEC 61800-2, IEC 61800-3, IEC 61800-5-1, IEC 61800-5-2, IEC 61800-6, IEC 60146-1-1, IEC 60146-1-2, IEC 61000-4-2, IEC 61000 4-4, IEC 61000-4-5, IEC 61000-4-7, IEC 61000-4-30, IEC 62477-1, IEC 61000-2-4, IEC 61800-9-1, IEC/TR 60146		Comply with IEC 60204-1 IEC 61800-2 IEC 61800-3 IEC 61800-5-1 IEEE519
1.9.	Units of measure	SI		
2.	Scope			
2.1.	Engineering	Yes		Comply
2.2.	Manufacturing	Yes		Comply
2.3.	Contractual documentation to be submitted (see 7043-SPE-COX-950-734-0001_ LV Variable frequency drive TS)	Yes		Noted
2.4.	Supply	Yes		Comply
2.5.	Factory tests	Yes		Comply
2.6.	Transport to site	Yes		Noted
2.7.	Unloading	Not applicable		NA
2.8.	Assembly	Optional		NA
2.9.	Assembly supervision	Yes		Comply
2.10.	Commissioning	Optional		Comply
2.11.	Commissioning supervision	Yes		Comply
2.12.	Spare parts for:			
	Testing and commissioning	Yes		Comply
	Three years of operation and mantenance	Yes		Comply
2.13.	Special tools required for operation and maintenance	Yes - if applicable		Noted
2.14.	Training program for operation and maintenance	Optional		Noted
2.15.	Required software and any special cable	Yes		Comply with SoMove

3.	**Site conditions (2)**			
3.1.	Site coordinates	Latitud: 31.106° N | Longitud: 27.902° E		**Noted**
3.2.	Location	Ras El-Hekma, Egypt		**Noted**
3.3.	**Atmospheric conditions:**			
	Maximum ambient temperature [ºC]	40ºC		**Comply**
	Minimum ambient temperature [ºC]	0ºC		**Comply**
	Altitude [m]	<1000		**Comply**
	Relative humidity [%]	95%		**Comply**
	Seismic level	Moderate (Zone 2, ECP 201)		**NA**

	Pollution level (IEC 60664-1)	-		2 conforming to IEC 61800-5-1
3.4.	**Indoor climatic conditions:**			
	Maximum ambient temperature [ºC] - Air conditioned rooms	40ºC		Comply
	Maximum ambient temperature [ºC] - Not in air conditioned rooms	40ºC		Comply
	Minimum ambient temperature [ºC]	0ºC		Comply
	Relative humidity [%]	95%		Comply without condensation conforming to IEC 60068-2-3
3.5.	**Design temperatures for VFD rating:**			
	Indoors [ºC]	40ºC		Comply
	Maximum operation temperature (without derating)	40ºC		Comply
	Maximum operation temperature (without derating)	(*)		40 C
	Output current derating vs operation temperature factor (%/ºC)	(*)		2 % as per the derating curve
	Outdoors [ºC]	Not applicable		NA
3.6.	**Classification of environmental conditions (IEC 60721-3):**			
	Storage	1K2/ 1B2/ 1S1/1M1		Ambient air temperature for storage -25…70 °C
	Transportation	2K7/ 2B2/ 2S1/2M4		Transport temperature -25...+70 °C
	Stationary use at weather protected locations	3M1		Noted
	Stationary use at non-weather protected locations	Not applicable		NA
3.7.	Transfromer designed and sized according to all site conditions	(*)		
4.	**Hazardous area**			
4.1	Hazardous area classification (IEC 60079)	Not applicable		NA
5.	**Voltage supply and network impedance**			
5.1	VFD Power supply	400 V ± 10%		Comply
5.2	Available auxiliaries power supplies(2) LV a.c. (non UPS)(energy consumed (VA)) UPS (only for control)(energy consumed (VA)) DC (only for control)(energy consumed (VA))	400/230 Vac±10% 230 Vac±10% 125 Vdc+10%-15%		Comply except 125Vdc
5.3	Short circuit power at point of coupling, Scc [MVA]	(*)		Will be supplied later
5.4	Short circuit ratio, Rsc	(*)		50 kA (100 ms) with pre-fuse or circuit breaker
5.5	Overvoltage category to IEC 61800-5.	Overvoltage Category 3		Category III according to EN 50178
6	**Constructive design**			
6.1	Installation	Indoors		Noted
6.2	Type	cabinet-built		Comply
6.3	Protection degree of the enclosure	IP-41		IP54
6.4	**Cooling system :**			
	Type	Air		Air cooled with forced

				ventilation
	Fans (nº x %)	(*)		1 Fan , Power part 1160 m3/h, Control part 140 m3/h
	Air flow [m3/h]	(*)		
	Pumps (nº x %)	(*)		
	Water consumption [l/s]	Not applicable		NA
	Water pressure acceptable at inlet	Not applicable		NA
	Heat dissipation [kW]	(*)		6.4 KW
6.5	Sound power level, L_W [dB (A) at 1 m] Guaranteed value	(*)		70 dB(A)
6.6	Mean time between failures (MTBF) [h] Guaranteed value	(*)		MTBF(Continuous Operation 76 000 h) & Intermittent Operation 103 000 h
6.7	Mean time to repair (MTTR) [h] Guaranteed value	(*)		
6.8	VFD life [years] (**)	(*)		Up to 20 years
6.10	Flanges to connect directly to an external exhaust air duct	(*)		Comply
6.11	**Painting**			
6.11.1	Finishing	RAL 7035		Light grey (RAL 7035)
6.11.2	Category to ISO 12944	C5-M		Comply
6.12	**Dimensions (including cooling fans)**			
	Length [mm]	(*)		600 mm
	Wide [mm]	(*)		600 mm
	Height [mm] - panel plinth only	(*)		
	Height [mm] - total including panel plinth	(*)		2350 mm
6.13	Weight [kg]	(*)		
6.14	**Required clearances for access during operation and maintenance**			
	Front [mm]	(*)		➢ 1000 mm
	Back [mm]	(*)		➢ 0 mm
	Side [mm]	(*)		Side mounting is accepted
	Top [mm]	(*)		➢ 100 mm
7.	Electrical design			

7.1.	Input:			
	Maximum short-circuit withstand current, [kA]	65 kA		**Applicable with External CB or fuses**
	Maximum short-circuit withstand duration [s]	1 s		**Comply with External CB or fuses**
	Short-circuit contribution multiplying factor for a fault at input terminal [% IN]	(*)		**NA**
	Rated system voltage, ULN [V]	400 V ± 10%		**Comply**
	Minimum voltage required for starting [% UN]	0,9		**Comply**
	Rated system frequency, ¦LN [Hz]	50Hz ± 5%		**Comply**
	Rated power, PN [kW]	160 kW		**Comply**
	Line-side rated a.c. current, ILN [A]	249.03 A		**252 A at 400 V (normal duty)**
	Fundamental line side rated a.c. current , ILN1 [A]	(*)		**1.1 x In during 60 s (normal duty)**
	Maximum current absorbed by VFD, considering the maximum power required in driven machine and the efficiency of all the equipment [% IN]	(*)		
	Power factor	0,95 Minimum		**99 %**
	Power factor at no load	(*)		**(from 30 % Pn) the ATV680 leads a sinusoidal mains current with a power factor cos Phi ≈ 1**
	Line-side displacement factor [cos jL1]	(*)		
	Supported neutral earthing schemes	(*)		
7.2.	**Output:**			
	Short-circuit contribution multiplying factor for a fault at output terminal [% IaN]	(*)		**Comply**
	Load-side converter a.c. rated voltage, UaN1 [V]	400 V		
	Maximum voltage variation [%]	0,1		
	Operating frequency range	(*)		**0.1…500 Hz**
	Maximum active power available at design temperature [kW]	(*)		**Un = 400 V 174 kVA , Rated output current In 302 A , Deration factor is 2 % for each degree C**
	Rated continuous output current, IaN [A]	(*)		
	Rated fundamental output current, IaN1 [A]	(*)		
	Maximum active power available at maximum temperature with derating [kW]	(*)		
	Rated continuous output current, at maximum temperature with derating IaN [A]	(*)		
	Rated fundamental output current at maximum temperature with derating, IaN1 [A]	(*)		
	Overload capability , IaM [A] Duration, [s]	(*)		**1.1 x In during 60 s (normal duty)**
	Rated load power factor	(*)		**Up to power factor cos Phi ≈ 1**
7.3.	Number of phases	3		**Comply**
7.4.	Efficiency, hC [%]	97%		**Up to 98 %**
	hC (25%)	(*)		
	hC (50%)	(*)		
	hC (75%)	(*)		
	hC (100%)	(*)		
7.5.	Maximum guaranteed losses at rated power [kW] Guaranteed value	(*)		**2 %**
7.6.	Regeneration system	(*)		**Not applicable on ATV680**
7.7.	Speed control system	Sensorless closed loop		**Comply**
7.8.	Low voltage VFD: Overvoltage category according to IEC 60664	Not applicable		**Category III according to EN 50178**
7.9.	Earthing system	TT		**6 pulse supply / TN-C or TN-S mains without neutral**

7.1.	**Rated insulation level**			
	Switchboard and bus bars, Ui [kV]	0,6 kV		Comply
	Switching devices	0,4 kV		Comply
	Wiring:			
	Power circuits	400 V		Comply
	Control circuits	230 V		Comply
	LV AC cables color coding and labeling according IEC 60446	L1 – Brown		As per factory standard
		L2 – Black		As per factory standard
		L3 – Grey		As per factory standard
		N - Blue		As per factory standard
		PE – yellow-green		As per factory standard
	DC power supply cables color coding and labeling according IEC 60446	+		As per factory standard
		–		As per factory standard
8.	**Power semiconductor**			
8.1.	**Rectifier**			
	Type (Diode, thyristor, IGBT, etc.)	(*)		IGBT
	Total number of pulses	AFE		Comply
	Rated current	(*)		252 A
	Rated voltage	(*)		400 V
	Duty class	(*)		ND

8.2	DC Bus (type)	(*)		DC capacitors
8.3	**Inverter+**			
	Type (Diode, thyristor, IGBT, etc.)	(*)		IGBT
	Total number of pulses	AFE		6
	Rated current	(*)		302
	Rated voltage	(*)		400
	Duty class	(*)		ND
9.	**EMC, input and output filters**			
9.1	Environmental category (IEC 61800-3)	Second		Integrated conforming to IEC 61800-3, category C3, shielded cable with 50 m Integrated conforming to IEC 61800-3, category C4, unshielded cable with 80 m
9.2	Emission category (IEC 61800-3)	C3		Comply
9.3	Input Filters	(*)		Comply
9.4	Output Filters	(*)		Comply with du/dt filter of 150 mt cable
9.4.1	Output dv / dt	(*)		Comply
9.4.2	Output common mode filter	(*)		NA
9.4.3	Output sinus filter	(*)		NA
9.5	Output dv / dt	(*)		Comply
9.6	Inverter commutating frequency	(*)		2.5 KHZ
9.7	All necessary filters are located within the same equipment	Yes		Yes
10.	**Harmonics according to Rsc value of item 5.4**			
10.1	**THDi (Input)**			
	At rated power	(*)		THDi is less than 2 %
	At ¾ rated power	(*)		
	At ½ rated power	(*)		
	At ¼ rated power	(*)		
10.2	**THDi (Output)**			
	At rated power	(*)		THDi<5 %
	At ¾ rated power	(*)		
	At ½ rated power	(*)		
	At ¼ rated power	(*)		
10.3	**Line side injected harmonics at rated power**			
	3rd (%In)	(*)		Comply with IEEE519 with AFE technology
	5th (%In)	(*)		
	7th (%In)	(*)		
	9th (%In)	(*)		
	9th (%In)	(*)		
	11th (%In)	(*)		
	13th (%In)	(*)		
	15th (%In)	(*)		
	17th (%In)	(*)		
	19th (%In)	(*)		
	21st (%In)	(*)		
	23rd (%In)	(*)		
	25th (%In)	(*)		

	27th (%In)	(*)		
	29th (%In)	(*)		
	31th (%In)	(*)		
	33th (%In)	(*)		
	35th (%In)	(*)		
	37th (%In)	(*)		
	39th (%In)	(*)		
	47th (%In)	(*)		
11.	**Protections**			
11.1	50/51 Phase over current	Yes		**Yes**
11.2	27 Under voltage (L-L or L-N)	Yes		**Yes**
11.3	51LR Locked rotor > 100 kW	Yes		**Yes**
11.4	49RMS Thermal overload monitoring by PT100 inputs	Yes - Eight (8) 3-wire, Pt-100 inputs per VFD		**Yes with 8 PT100 relays**
11.5	66 Maximum number of starting (if it is applicable)	Yes		**Yes**
11.6	46 Negative-sequence overcurrent	Yes		**Yes**
11.7	49 Thermal relay	Yes		**Yes**

11.8	51N Sensitive Earth fault detection	Yes		Yes
12.	Auxiliaries(3)			
12.1	Brake resistance	(*)		NA , NA , Comply , Comply with feeder for motor heater
12.2	Energy absorption rating of brake resistance	(*)		
12.3	Re-start after voltage drop of 1 second	Yes		
12.4	Motor heater powering and control	Yes		
12.5	Total power consumption of all auxiliaries including cooling system [kW]	(*)		
12.6	Tripping coil to emergency stop	Yes - EPB in front		Yes with Emergency stop push button according SIL 3 stop category 0
12.7	Safety Integrity Level (SIL)			
	STO (Safe torque off)	Not Applicable		Comply
	SS1 (safe Stop 1)	Not Applicable		NA
	SS2 (safe Stop 2)	Not Applicable		NA
	Other safety functions	Not Applicable		NA
	Emergency stop button Category Opening breaker	No		Comply
	Safety relay with feedback	Not Applicable		NA
12.8	Ride-through mode			
	Voltage sags down to 80% of the VFD rated voltage	≥ 5 seconds		-10 % of nominal voltage Starting the drive and continuous operation possible (1) , -15 % of nominal voltage Starting the drive and operation (1) for 10 s per 100 s possible -20 , -20 % of nominal voltage Operation (1) for less than 1 s possible , -30 % of nominal voltage Operation (1) for less than 0.5 s possible Comply with catch on fly option
	Voltage sags down to 70% of the VFD rated voltage	≥ 1 second		
	Voltage sags down to 40% of the VFD rated voltage	≥ 0,2 seconds		
	Voltage sags down to 0% of the VFD rated voltage	≥ 0,1 seconds		
12.9	Motor flying start	Yes		
12.10	Internal equipment			
	Input on-load switch	Yes		Comply
	High-speed fuses	Yes		Comply
	Line contactor	Yes		No need
	Heater	Yes - Hygrostatically controlled		Comply
	Lighting	Yes		Comply
	Power socket	Yes		Comply
12.11	Spare availability			
	Year of model start of manufacturing	(*)		Since 2018
	Spares availability after End of Life of the model	(*)		15 to 20 years
13.	Control and instrumentation			

13.1	Communication protocol	Modbus TCP/IP		Comply
13.2	**Digital inputs/ output**			
	Start / Stop input	Yes		Comply
	Start / Stop output	Yes		Comply
	Forward / Reverse output	Yes		Comply
	Acceleration / deceleration output	Yes		Comply
	Internal failure output	Yes		Comply
	External trip output	Yes		Comply
	Local / remote output	Yes		Comply
13.3	**Analog inputs / outputs**			
	Current output	Yes		Comply
	Motor speed output	Yes		Comply
	Speed reference input	Yes		Comply
	Voltage output	Yes		Comply
	Power output	Yes		Comply
	Torque output	Yes		Comply
	Motor frequency	Yes		Comply
	Power factor at the motor	Yes		Comply
	IGBT temperature	Yes		Comply
	Galvanic insulation for analog signals	(*)		Comply
14.	**Power cables requirements**			
14.1	**Cables from incoming to VFD**			
	Conductor	Cu		Ok
	Configuration (3-pole/single-pole)	3 pole symetrical conductors for protective grounding (delta configuration)(Note (6))		1x (3x 185 mm²) or 2x (3x 95 mm²)
	Magnetic shield	Yes		No need for Sheild
	Max cross section	(*)		

	Max number per phase	(*)		As per Cable supplier information's
	Insulation level	(*)		
	Maximum length (screened cable) without filters	(*)		
	Maximum length (non-screened cable) without filters	(*)		
	Special requirement for PE conductor	(*)		
	Special requirement when motor is in hazardous area	(*)		
	Other requisites	-		
14.2.	**Cables from to motor**			
	Conductor	Cu		Ok
	Configuration (3-pole/single-pole)	3 pole symetrical conductors for protective grounding (delta configuration)(Note (6))		Ok
	Magnetic shield	Yes		2x (3x 240 mm²) or 3x (3x 120 mm²) or 4x (3x 95 mm²)
	Max cross section	(*)		
	Max number per phase	(*)		
	Insulation level	(*)		
	Maximum length (screened cable) without filters	(*)		
	Maximum length (non-screened cable) without filters	(*)		
	Special requirement for PE conductor	(*)		
	Special requirement when motor is in hazardous area	(*)		
	Other requisites	(*)		
14.3.	**Cables from Auxiliary supply to VFD**			
	Conductor	Cu		Ok
	Configuration	3-pole		
	Magnetic shield	No		
	Max cross section	(*)		
	Max number per phase	(*)		
	Insulation level	(*)		
	Special requirement for PE conductor	(*)		
	Other requisites	(*)		
14.4.	**Control Cables**			
	Number of wires	(*)		Comply
	Magnetic shield	Yes		
	Other requisites	(*)		
	Power consumption	(*)		
15.	**Tests**			
15.1.	**Routine tests**			
	Insulation test (IEC 60146 and 61800-5)	Yes		Ok
	Light load and functional test	Yes		Ok
	Rated current test	Yes		Ok
	Checking of auxiliary devices	Yes		Ok
	Checking the properties of the control equipment	Yes		Ok
	Checking the protective devices	Yes		Ok
15.2.	**Type tests**			
	Rated current	Include test certificates		Ok
	Power loss	Include test certificates		Ok
	Temperature rise test	Include test certificates		Ok
	EM immunity test	Include test certificates		Ok
	EM emission test	Include test certificates		Ok
	Current sharing	Include test certificates		Ok
	Voltage division	Include test certificates		Ok
15.3.	**Special tests**			

	Over current capability test	Optional - to be quoted separately		NA
	Measurement of inherent voltage regulation	Optional - to be quoted separately		NA
	Measurement of ripple voltage and current	Optional - to be quoted separately		NA
	Measurement of harmonic currents	Optional - to be quoted separately		NA
	Radio frequency radiated and conducted disturbances	Optional - to be quoted separately		NA

	Power factor measurement	Yes - with oscilloscope to display the VFD input wave		NA
	Load tests	Yes - one (1) VFD only		Ok
	Noise test	Yes - one (1) VFD only		NA
	RideThrough test	The Bidder will provide in any case copies of the records to certificate that the VF complies with IEC 61000-4-34 or alternatively with SEMI F47		NA
15.4.	**Power drive system test (special test)**			
	Light load	Not Applicable		As per FAT
	Load	Not Applicable		As per FAT
	Load duty	Not Applicable		As per FAT
	Allowance FLC (full load current) vs speed	Not Applicable		As per FAT
	Temperature rise	Not Applicable		As per FAT
	Efficiency	Not Applicable		As per FAT
	Line-side current harmonic content	Not Applicable		As per FAT
	Power factor	Not Applicable		As per FAT
	Current sharing	Not Applicable		As per FAT
	Voltage division	Not Applicable		As per FAT
	Checking of auxiliary devices	Not Applicable		As per FAT
	Checking coordination of protective devices	Not Applicable		As per FAT
	Checking properties under unusual service conditions	Not Applicable		As per FAT
	Shaft current / bearing insulation	Not Applicable		As per FAT
	Audible noise	Not Applicable		As per FAT
	Motor vibration	Not Applicable		As per FAT
	EMC	Not Applicable		As per FAT
	Harmonic content of the complete drive module output current	Not Applicable		As per FAT
	Current limit and current loop	Not Applicable		As per FAT
	Speed loop	Not Applicable		As per FAT
	Torque pulsation	Not Applicable		As per FAT
	Automatic restart	Not Applicable		As per FAT
16.	**Documents and drawings**			
16.1	Deviations from specification(2)	(*)		Ok
16.2	Seismic qualification test certificates/calculations	(*)		NA
16.3	**Spare parts(3)**			
	For testing and commissioning	Yes		Ok
	For three years of operation (optional)	Yes		Ok
16.3	Equipment layout with overall dimensions, weight, clearances, cable entries, fixing bolts and supports	Yes		Ok
16.4	Efficiency losses, PF curves from no-load to 150% rated load	Yes		NA
16.5	Electrical wiring diagrams (indicating terminal codifications)	Yes		Ok
16.6	Operation and maintenance manuals	Yes		OK
16.7.	VFD sizing criteria according to motor and driven equipment	Yes		Ok
16.8	Tropicalization, or any other special coating procedure	Yes		Ok

16.9.	Signal list	Yes		Ok
16.10.	Installation guide to comply with SIL category	Not Applicable		Ok
16.11	Cabinet heat dissipation	Yes		Ok
16.12.	Detailed planning	Yes		Ok
16.13.	Quality control plan	Yes		Ok
16.14.	**Detailed performance test procedures showing test schedule and techniques and including at least:**			
	• Name of item to be tested.	Yes		As per FAT
	• Location of measurement points.	Yes		As per FAT
	• Manufacturer and model number of instrumentation used.	Yes		As per FAT
	• Any applicable correction curve.	Yes		As per FAT
	• Verifiable method of the test result calculation.	Yes		As per FAT
	Method to correct test conditions back to design conditions.	Yes		As per FAT
16.15.	Tests and certificates	Yes		Ok
16.16.	Final dossier	Yes		NA
16.17.	PPI Monthly report	Yes		NA

MODON ORASCOM CONSTRUCTION TROJAN COX

LV frecuency converter DSH (Washing Blowers)

Item	Description	Requested data	Offer	Offered data (*)
1.	General			
1.1.	Supplier	(*)		Schneider Electric
1.2.	Catalogue No.	(*)		
1.3.	KKS code	TBD		
1.4.	Manufacturer	(*)		Schneider Electric
1.5.	Place of manufacture	(*)		
1.6.	Year of manufacture	(*)		2026 or 2027
1.7.	Quantity	2		
1.8.	Applicable standards	IEC 61800-2, IEC 61800-3, IEC 61800-5-1, IEC 61800-5-2, IEC 61800-6, IEC 60146-1-1, IEC 60146-1-2, IEC 61000-4-2, IEC 61000 4-4, IEC 61000-4-5, IEC 61000-4-7, IEC 61000-4-30, IEC 62477-1, IEC 61000-2-4, IEC 61800-9-1, IEC/TR 60146		Comply with IEC 60204-1 IEC 61800-2 IEC 61800-3 IEC 61800-5-1 IEEE519
1.9.	Units of measure	SI		
2.	Scope			
2.1.	Engineering	Yes		Comply
2.2.	Manufacturing	Yes		Comply
2.3.	Contractual documentation to be submitted (see 7043-SPE-COX-950-734-0001_ LV Variable frequency drive TS)	Yes		Noted
2.4.	Supply	Yes		Comply
2.5.	Factory tests	Yes		Comply
2.6.	Transport to site	Yes		Noted
2.7.	Unloading	Not applicable		NA
2.8.	Assembly	Optional		NA
2.9.	Assembly supervision	Yes		Comply
2.10.	Commissioning	Optional		Comply
2.11.	Commissioning supervision	Yes		Comply
2.12.	Spare parts for:			
	Testing and commissioning	Yes		Comply
	Three years of operation and mantenance	Yes		Comply
2.13.	Special tools required for operation and maintenance	Yes - if applicable		Noted
2.14.	Training program for operation and maintenance	Yes		Noted
2.15.	Required software and any special cable	Yes		Comply with SoMove
3.	Site conditions(2)			

3.1.	Site coordinates	Latitud: 31.106° N | Longitud: 27.902° E		Noted
3.2.	Location	Ras El-Hekma, Egypt		Noted
3.3.	Atmospheric conditions:			
	Maximum ambient temperature [ºC]	40ºC		Comply
	Minimum ambient temperature [ºC]	0ºC		Comply
	Altitude [m]	<1000		Comply
	Relative humidity [%]	95%		Comply
	Seismic level	Moderate (Zone 2, ECP 201)		NA
	Pollution level (IEC 60664-1)	-		2 conforming to IEC 61800-5-1
3.4.	Indoor climatic conditions:			
	Maximum ambient temperature [ºC] - Air conditioned rooms	40ºC		Comply

	Maximum ambient temperature [ºC] - Not in air conditioned rooms	40ºC		Comply
	Minimum ambient temperature [ºC]	0ºC		Comply
	Relative humidity [%]	95%		Comply without condensation conforming to IEC 60068-2-3
3.5.	Design temperatures for VFD rating:			
	Indoors [ºC]	40ºC		Comply
	Maximum operation temperature (without derating)	40ºC		Comply
	Maximum operation temperature (without derating)	(*)		40 C
	Output current derating vs operation temperature factor (%/ºC)	(*)		2 % as per the derating curve
	Outdoors [ºC]	Not applicable		NA
3.6.	Classification of environmental conditions (IEC 60721-3):			
	Storage	1K2/ 1B2/ 1S1/1M1		Ambient air temperature for storage -25…70 °C
	Transportation	2K7/ 2B2/ 2S1/2M4		Transport temperature -25...+70 °C
	Stationary use at weather protected locations	3M1		Noted
	Stationary use at non-weather protected locations	Not applicable		NA
3.7.	Transformer designed and sized according to all site conditions	(*)		
4.	Hazardous area			
4.1	Hazardous area classification (IEC 60079)	Not applicable		NA
5.	Voltage supply and network impedance			
5.1	VFD Power supply	400 V ± 10%		Comply
5.2	Available auxiliaries power supplies(2) LV a.c. (non UPS)(energy consumed (VA)) UPS (only for control)(energy consumed (VA)) DC (only for control)(energy consumed (VA))	400/230 Vac±10% 230 Vac±10% 125 Vdc+10%-15%		Comply except 125Vdc
5.3	Short circuit power at point of coupling, Scc [MVA]	(*)		Will be supplied later
5.4	Short circuit ratio, Rsc	(*)		50 kA (100 ms) with pre-fuse or circuit breaker
5.5	Overvoltage category to IEC 61800-5.	Overvoltage Category 3		Category III according to EN 50178
6	Constructive design			
6.1	Installation	Indoors		Noted
6.2	Type	cabinet-built		Comply
6.3	Protection degree of the enclosure	IP-41		IP54

6.4	Cooling system :			
	Type	Air		Air cooled with forced ventilation
	Fans (nº x %)	(*)		1 Fan , Power part 1160 m3/h, Control part 140 m3/h
	Air flow [m3/h]	(*)		
	Pumps (nº x %)	(*)		
	Water consumption [l/s]	Not applicable		NA
	Water pressure acceptable at inlet	Not applicable		NA
	Heat dissipation [kW]	(*)		6.4 KW
6.5	Sound power level, L_W [dB (A) at 1 m] Guaranteed value	<85db		70 dB(A)
6.6	Mean time between failures (MTBF) [h] Guaranteed value	(*)		MTBF(Continuous Operation 76 000 h) & Intermittent Operation 103 000 h
6.7	Mean time to repair (MTTR) [h] Guaranteed value	(*)		
6.8	VFD life [years] (**)	25		Up to 20 years
6.10	Flanges to connect directly to an external exhaust air duct	(*)		Comply
6.11	Painting			
6.11.1	Finishing	RAL 7035		Light grey (RAL 7035)
6.11.2	Category to ISO 12944	C5-M		Comply
6.12	Dimensions (including cooling fans)			
	Length [mm]	(*)		600 mm
	Wide [mm]	(*)		600 mm
	Height [mm] - panel plinth only	(*)		
	Height [mm] - total including panel plinth	(*)		2350 mm
6.13	Weight [kg]	(*)		
6.14	Required clearances for access during operation and maintenance			
	Front [mm]	(*)		1000 mm
	Back [mm]	(*)		0 mm
	Side [mm]	(*)		Side mounting is accepted
	Top [mm]	(*)		100 mm
7.	Electrical design			
7.1.	Input:			
	Maximum short-circuit withstand current, [kA]	65 kA		Applicable with External CB or fuses

	Maximum short-circuit withstand duration [s]	1 s		Comply with External CB or fuses
	Short-circuit contribution multiplying factor for a fault at input terminal [% IN]	(*)		NA
	Rated system voltage, ULN [V]	400 V ± 10%		Comply
	Minimum voltage required for starting [% UN]	0,9		Comply
	Rated system frequency, ¦LN [Hz]	50Hz ± 5%		Comply
	Rated power, PN [kW]	160 kW		Comply
	Line-side rated a.c. current, ILN [A]	259.94 A		252 A at 400 V (normal duty)
	Fundamental line side rated a.c. current , ILN1 [A]	(*)		1.1 x In during 60 s (normal duty)
	Maximum current absorbed by VFD, considering the maximum power required in driven machine and the efficiency of all the equipment [% IN]	(*)		
	Power factor	0,95 minmum		99 %
	Power factor at no load	(*)		(from 30 % Pn) the ATV680 leads a sinusoidal mains current with a power factor cos Phi ≈ 1
	Line-side displacement factor [cos jL1]	(*)		
	Supported neutral earthing schemes	(*)		
7.2.	Output:			
	Short-circuit contribution multiplying factor for a fault at output terminal [% IaN]	(*)		Comply
	Load-side converter a.c. rated voltage, UaN1 [V]	400 V		
	Maximum voltage variation [%]	0,1		
	Operating frequency range	(*)		0.1…500 Hz
	Maximum active power available at design temperature [kW]	(*)		Un = 400 V 174 kVA , Rated output current In 302 A , Deration factor is 2 % for each degree C
	Rated continuous output current, IaN [A]	(*)		
	Rated fundamental output current, IaN1 [A]	(*)		
	Maximum active power available at maximum temperature with derating [kW]	(*)		
	Rated continuous output current, at maximum temperature with derating IaN [A]	(*)		
	Rated fundamental output current at maximum temperature with derating, IaN1 [A]	(*)		
	Overload capability , IaM [A] Duration, [s]	(*)		1.1 x In during 60 s (normal duty)
	Rated load power factor	(*)		Up to power factor cos Phi ≈ 1
7.3.	Number of phases	3		Comply
7.4.	Efficiency, [%]	97%		comply
	hC(25%)			
	hC(50%)			
	hC(75%)			
	hC(100%)			
7.5.	Maximum guaranteed losses at rated power [kW] Guaranteed value	(*)		3%
7.6.	Regeneration system	(*)		Not applicable on

				ATV680
7.7.	Speed control system	Sensorless closed loop		Comply
7.8.	Rated insulation level			
	Switchboard and bus bars, Ui [kV]	0,6 kV		Comply
	Switching devices	0,4 kV		Comply
	Wiring:			
	Power circuits	400 V		Comply
	Control circuits	230 V		Comply
	LV AC cables color coding and labeling according IEC 60446	L1 – Brown		As per factory standard
		L2 – Black		As per factory standard
		L3 – Grey		As per factory standard
		N - Blue		As per factory standard
		PE – yellow-green		As per factory standard
	DC power supply cables color coding and labeling according IEC 60446	+		As per factory standard
		–		As per factory standard
8.	Power semiconductor			
8.1.	Rectifier			
	Type (Diode, thyristor, IGBT, etc.)	(*)		IGBT
	Total number of pulses	AFE		Comply
	Rated current	(*)		252 A
	Rated voltage	(*)		400 V
	Duty class	(*)		ND

8.2	DC Bus (type)	(*)		DC Capacitors
8.3	Inverter+			
	Type (Diode, thyristor, IGBT, etc.)	(*)		IGBT
	Total number of pulses	AFE		6
	Rated current	(*)		302
	Rated voltage	(*)		400
	Duty class	(*)		ND
9.	EMC, input and output filters			
9.1	Environmental category (IEC 61800-3)	Second		Integrated conforming to IEC 61800-3, category C3, shielded cable with 50 m Integrated conforming to IEC 61800-3, category C4, unshielded cable with 80 m
9.2	Emission category (IEC 61800-3)	C3		Comply
9.3	Input Filters	(*)		Comply
9.4	Output Filters	(*)		Comply with du/dt filter of 150 mt cable
9.4.1	Output dv / dt	(*)		Comply
9.4.2	Output common mode filter	(*)		NA
9.4.3	Output sinus filter	(*)		NA
9.5	Output dv / dt	(*)		Comply
9.6	Inverter commutating frequency	(*)		2.5 KHZ
9.7	All necessary filters are located within the same equipment	Yes		Yes
10.	Harmonics according to Rsc value of item 5.4			
10.1	THDi (Input)			
	At rated power	(*)		THDi is less than 2 %
	At ¾ rated power	(*)		
	At ½ rated power	(*)		
	At ¼ rated power	(*)		
10.2	THDi (Output)			
	At rated power	(*)		THDi<5 %
	At ¾ rated power	(*)		
	At ½ rated power	(*)		
	At ¼ rated power	(*)		
10.3	Line side injected harmonics at rated power			
	3rd (%In)	(*)		Comply with IEEE519 with
	5th (%In)	(*)		
	7th (%In)	(*)		
	9th (%In)	(*)		

	9th (%In)	(*)		AFE technology
	11th (%In)	(*)		
	13th (%In)	(*)		
	15th (%In)	(*)		
	17th (%In)	(*)		
	19th (%In)	(*)		
	21st (%In)	(*)		
	23rd (%In)	(*)		
	25th (%In)	(*)		
	27th (%In)	(*)		
	29th (%In)	(*)		
	31th (%In)	(*)		
	33th (%In)	(*)		
	35th (%In)	(*)		
	37th (%In)	(*)		
	39th (%In)	(*)		
	47th (%In)	(*)		
11.	Protections			
11.1	50/51 Phase over current	Yes		Yes
11.2	27 Under voltage (L-L or L-N)	Yes		Yes
11.3	51LR Locked rotor > 100 kW	Yes		Yes
11.4	49RMS Thermal overload monitoring by PT100 inputs	Yes - Eight (8) 3-wire, Pt-100 inputs per VFD		Yes with 8 PT100 relays
11.5	66 Maximum number of starting (if it is applicable)	Yes		Yes
11.6	46 Negative-sequence overcurrent	Yes		Yes

11.7	49 Thermal relay	Yes		Yes
11.8	51N Sensitive Earth fault detection	Yes		Yes
12.	Auxiliaries(3)			
12.1	Brake resistance	(*)		NA , NA , Comply , Comply with feeder for motor heater
12.2	Energy absorption rating of brake resistance	(*)		
12.3	Re-start after voltage drop of 1 second	Yes		
12.4	Motor heater powering and control	Yes		
12.5	Total power consumption of all auxiliaries including cooling system [kW]	(*)		
12.6	Tripping coil to emergency stop	Yes - EPB in front		Yes with Emergency stop push button according SIL 3 stop category 0
12.7	Safety Integrity Level (SIL)			
	STO (Safe torque off)	Not Applicable		Comply
	SS1 (safe Stop 1)	Not Applicable		NA
	SS2 (safe Stop 2)	Not Applicable		NA
	Other safety functions	Not Applicable		NA
	Emergency stop button Category Opening breaker	No		Comply
	Safety relay with feedback	Not Applicable		NA
12.8	Ride-through mode			
	Voltage sags down to 80% of the VFD rated voltage	≥ 5 seconds		-10 % of nominal voltage Starting the drive and continuous operation possible (1) , -15 % of nominal voltage Starting the drive and operation (1) for 10 s per 100 s possible -20 , -20 % of nominal voltage Operation (1) for less than 1 s possible , -30 % of nominal voltage Operation (1) for less than 0.5 s possible Comply with catch on fly option
	Voltage sags down to 70% of the VFD rated voltage	≥ 1 second		
	Voltage sags down to 40% of the VFD rated voltage	≥ 0,2 seconds		
	Voltage sags down to 0% of the VFD rated voltage	≥ 0,1 seconds		
12.9	Motor flying start	Yes		
12.10	Internal equipment			
	Input on-load switch	Yes		Comply
	High-speed fuses	Yes		Comply
	Line contactor	Yes		No need
	Heater	Yes - Hygrostatically controlled		Comply
	Lighting	Yes		Comply
	Power socket	Yes		Comply

12.11	Spare availability			
	Year of model start of manufacturing	(*)		Since 2018
	Spares availability after End of Life of the model	(*)		15 to 20 years
13.	Control and instrumentation			
13.1	Communication protocol	Modbus TCP/IP		Comply
13.2	Digital inputs/ output			
	Start / Stop input	Yes		Comply
	Start / Stop output	Yes		Comply
	Forward / Reverse output	Yes		Comply
	Acceleration / deceleration output	Yes		Comply
	Internal failure output	Yes		Comply
	External trip output	Yes		Comply
	Local / remote output	Yes		Comply
13.3	Analog inputs / outputs			
	Current output	Yes		Comply
	Motor speed output	Yes		Comply
	Speed reference input	Yes		Comply
	Voltage output	Yes		Comply
	Power output	Yes		Comply
	Torque output	Yes		Comply
	Motor frequency	Yes		Comply
	Power factor at the motor	Yes		Comply
	IGBT temperature	Yes		Comply
	Galvanic insulation for analog signals	(*)		Comply

14.	Power cables requirements			
14.1.	Cables from incoming to VFD			
	Conductor	Cu		Ok
	Configuration (3-pole/single-pole)	3 pole symetrical conductors for protective grounding (delta configuration)(Note (6))		1x (3x 185 mm²) or 2x (3x 95 mm²)
	Magnetic shield	Yes		No need for Sheild As per Cable supplier information's
	Max cross section	(*)		
	Max number per phase	(*)		
	Insulation level	(*)		
	Maximum length (screened cable) without filters	(*)		
	Maximum length (non-screened cable) without filters	(*)		
	Special requirement for PE conductor	(*)		
	Special requirement when motor is in hazardous area	(*)		
	Other requisites	-		
14.2.	Cables from VFD to motor			
	Conductor	Cu		Ok
	Configuration (3-pole/single-pole)	3 pole symetrical conductors for protective grounding (delta configuration)(Note (6))		Ok
	Magnetic shield	Yes		2x (3x 240 mm²) or 3x (3x 120 mm²) or 4x (3x 95 mm²)
	Max cross section	(*)		
	Max number per phase	(*)		
	Insulation level	(*)		
	Maximum length (screened cable) without filters	(*)		
	Maximum length (non-screened cable) without filters	(*)		
	Special requirement for PE conductor	(*)		
	Special requirement when motor is in hazardous area	(*)		
	Other requisites	(*)		
14.3.	Cables from Auxiliary supply to VFD			
	Conductor	Cu		Ok
	Configuration	3-pole		
	Magnetic shield	No		
	Max cross section	(*)		
	Max number per phase	(*)		
	Insulation level	(*)		
	Special requirement for PE conductor	(*)		
	Other requisites	(*)		
14.4.	Control Cables			
	Number of wires	(*)		comply
	Magnetic shield	Yes		
	Other requisites	(*)		
	Power consumption	(*)		
15.	Tests			
15.1.	Routine tests			
	Insulation test (IEC 60146 and 61800-5)	Yes		Ok
	Light load and functional test	Yes		Ok
	Rated current test	Yes		Ok
	Checking of auxiliary devices	Yes		Ok
	Checking the properties of the control equipment	Yes		Ok
	Checking the protective devices	Yes		Ok
15.2.	Type tests			
	Rated current	Include test certificates		Ok
	Power loss	Include test certificates		Ok
	Temperature rise test	Include test certificates		Ok

	EM immunity test	Include test certificates		Ok
	EM emission test	Include test certificates		Ok
	Current sharing	Include test certificates		Ok
	Voltage division	Include test certificates		Ok
15.3.	Special tests			

	Over current capability test	Optional - to be quoted separately		NA
	Measurement of inherent voltage regulation	Optional - to be quoted separately		NA
	Measurement of ripple voltage and current	Optional - to be quoted separately		NA
	Measurement of harmonic currents	Optional - to be quoted separately		NA
	Radio frequency radiated and conducted disturbances	Optional - to be quoted separately		NA
	Power factor measurement	Yes - with oscilloscope to display the VFD input wave		NA
	Load tests	Yes - one (1) VFD only		Ok
	Noise test	Yes - one (1) VFD only		NA
	RideThrough test	The Bidder will provide in any case copies of the records to certificate that the VF complies with IEC 61000-4-34 or alternatively with SEMI F47.		NA
15.4.	Power drive system test (special test)			
	Light load	Not Applicable		As per FAT
	Load	Not Applicable		As per FAT
	Load duty	Not Applicable		As per FAT
	Allowance FLC (full load current) vs speed	Not Applicable		As per FAT
	Temperature rise	Not Applicable		As per FAT
	Efficiency	Not Applicable		As per FAT
	Line-side current harmonic content	Not Applicable		As per FAT
	Power factor	Not Applicable		As per FAT
	Current sharing	Not Applicable		As per FAT
	Voltage division	Not Applicable		As per FAT
	Checking of auxiliary devices	Not Applicable		As per FAT
	Checking coordination of protective devices	Not Applicable		As per FAT
	Checking properties under unusual service conditions	Not Applicable		As per FAT
	Shaft current / bearing insulation	Not Applicable		As per FAT
	Audible noise	Not Applicable		As per FAT
	Motor vibration	Not Applicable		As per FAT
	EMC	Not Applicable		As per FAT
	Harmonic content of the complete drive module output current	Not Applicable		As per FAT
	Current limit and current loop	Not Applicable		As per FAT
	Speed loop	Not Applicable		As per FAT
	Torque pulsation	Not Applicable		As per FAT
	Automatic restart	Not Applicable		As per FAT
16.	Documents and drawings			
16.1.	Deviations from specification(2)	(*)		Ok
16.2.	Seismic qualification test certificates/calculations	(*)		NA
16.3.	Spare parts(3)			
	For testing and commissioning	Yes		Ok
	For three years of operation (optional)	Yes		Ok

16.3.	Equipment layout with overall dimensions, weight, clearances, cable entries, fixing bolts and supports	Yes		Ok
16.4.	Efficiency losses, PF curves from no-load to 150% rated load	Yes		NA
16.5.	Electrical wiring diagrams (indicating terminal codifications)	Yes		Ok
16.6.	Instalation,operation and maintenance manuals	Yes		OK
16.7.	VFD sizing criteria according to motor and driven equipment	Yes		Ok
16.8.	Tropicalization, or any other special coating procedure	Yes		Ok
16.9.	Signal list	Yes		Ok
16.10.	Installation guide to comply with SIL category	Not Applicable		Ok
16.11.	Cabinet heat dissipation	Yes		Ok
16.12.	Detailed planning	Yes		Ok
16.13.	Quality control plan	Yes		Ok

16.14.	Detailed performance test procedures showing test schedule and techniques and including at least:			
	• Name of item to be tested.	Yes		As per FAT
	• Location of measurement points.	Yes		As per FAT
	• Manufacturer and model number of instrumentation used.	Yes		As per FAT
	• Any applicable correction curve.	Yes		As per FAT
	• Verifiable method of the test result calculation.	Yes		As per FAT
	Method to correct test conditions back to design conditions.	Yes		As per FAT
16.15.	Tests and certificates	Yes		Ok
16.16.	Final dossier	Yes		NA
16.17.	PPI Monthly report	Yes		NA

MODON ORASCOM CONSTRUCTION TROJAN COX

LV frecuency converter DSH (Backwash pumps)

Item	Description	Requested data	Offer	Offered data (*)
1.	General			
1.1.	Supplier	(*)		Schneider Electric
1.2.	Catalogue No.	(*)		
1.3.	KKS code	TBD		
1.4.	Manufacturer	(*)		Schneider Electric
1.5.	Place of manufacture	(*)		
1.6.	Year of manufacture	(*)		2026 or 2027
1.7.	Quantity	2		
1.8.	Applicable standards	IEC 61800-2, IEC 61800-3, IEC 61800-5-1, IEC 61800-5-2, IEC 61800-6, IEC 60146-1-1, IEC 60146-1-2, IEC 61000-4-2, IEC 61000 4-4, IEC 61000-4-5, IEC 61000-4-7, IEC 61000-4-30, IEC 62477-1, IEC 61000-2-4, IEC 61800-9-1, IEC/TR 60146		Comply with IEC 60204-1 IEC 61800-2 IEC 61800-3 IEC 61800-5-1 IEEE519
1.9.	Units of measure	SI		
2.	Scope			
2.1.	Engineering	Yes		Comply
2.2.	Manufacturing	Yes		Comply
2.3.	Contractual documentation to be submitted (see 7043-SPE-COX-950-734-0001_ LV Variable frequency drive TS)	Yes		Noted
2.4.	Supply	Yes		Comply
2.5.	Factory tests	Yes		Comply
2.6.	Transport to site	Yes		Noted
2.7.	Unloading	Not applicable		NA
2.8.	Assembly	Optional		NA
2.9.	Assembly supervision	Yes		Comply
2.10.	Commissioning	Optional		Comply
2.11.	Commissioning supervision	Yes		Comply
2.12.	Spare parts for:			
	Testing and commissioning	Yes		Comply
	Three years of operation and mantenance	Yes		Comply
2.13.	Special tools required for operation and maintenance	Yes - if applicable		Noted
2.14.	Training program for operation and maintenance	Optional		Noted
2.15.	Required software and any special cable	Yes		Comply with SoMove
3.	Site conditions[2]			

3.1.	Site coordinates	Latitud: 31.106° N | Longitud: 27.902° E		Noted
3.2.	Location	Ras El-Hekma, Egypt		Noted
3.3.	Atmospheric conditions:			
	Maximum ambient temperature [ºC]	40ºC		Comply
	Minimum ambient temperature [ºC]	0ºC		Comply
	Altitude [m]	<1000		Comply
	Relative humidity [%]	95%		Comply
	Seismic level	Moderate (Zone 2, ECP 201)		NA
	Pollution level (IEC 60664-1)	-		2 conforming to IEC 61800-5-1
3.4.	Indoor climatic conditions:			

	Maximum ambient temperature [ºC] - Air conditioned rooms	40ºC		Comply
	Maximum ambient temperature [ºC] - Not in air conditioned rooms	40ºC		Comply
	Minimum ambient temperature [ºC]	0ºC		Comply
	Relative humidity [%]	95%		Comply without condensation conforming to IEC 60068-2-3
3.5.	Design temperatures for VFD rating:			
	Indoors [ºC]	40ºC		Comply
	Maximum operation temperature (without derating)	40ºC		Comply
	Maximum operation temperature (without derating)	(*)		40 C
	Output current derating vs operation temperature factor (%/ºC)	(*)		2 % as per the derating curve
	Outdoors [ºC]	Not applicable		NA
3.6.	Classification of environmental conditions (IEC 60721-3):			
	Storage	1K2/ 1B2/ 1S1/1M1		Ambient air temperature for storage -25…70 °C
	Transportation	2K7/ 2B2/ 2S1/2M4		Transport temperature -25...+70 °C
	Stationary use at weather protected locations	3M1		Noted
	Stationary use at non-weather protected locations	Not applicable		NA
3.7.	Transfromer designed and sized according to all site conditions	(*)		
4.	Hazardous area			
4.1	Hazardous area classification (IEC 60079)	Not applicable		NA
5.	Voltage supply and network impedance			
5.1	VFD Power supply	400 V ± 10%		Comply
5.2	Available auxiliaries power supplies(2) LV a.c. (non UPS)(energy consumed (VA)) UPS (only for control)(energy consumed (VA)) DC (only for control)(energy consumed (VA))	400/230 Vac±10% 230 Vac±10% 125 Vdc+10%-15%		Comply except 125Vdc
5.3	Short circuit power at point of coupling, Scc [MVA]	(*)		Will be supplied later
5.4	Short circuit ratio, Rsc	(*)		50 kA (100 ms) with pre-fuse or circuit breaker
5.5	Overvoltage category to IEC 61800-5.	Overvoltage Category 3		Category III according to EN 50178
6	Constructive design			
6.1	Installation	Indoors		Noted
6.2	Type	cabinet-built		Comply

6.3	Protection degree of the enclosure	IP-41		IP54
6.4	Cooling system :			
	Type	Air		Air cooled with forced ventilation
	Fans (nº x %)	(*)		Will be available on product drawing(Fan) , Powerpart 2320m3/h Controlpart 280m3/h
	Air flow [m3/h]	(*)		
	Pumps (nº x %)	(*)		
	Water consumption [l/s]	Not applicable		NA
	Water pressure acceptable at inlet	Not applicable		NA
	Heat dissipation [kW]	(*)		7890W
6.5	Sound power level, L_W [dB (A) at 1 m] Guaranteed value	(*)		73dB(A)
6.6	Mean time between failures (MTBF) [h] Guaranteed value	(*)		MTBF(Continuous Operation 82 000 h) & Intermittent Operation 98 000 h
6.7	Mean time to repair (MTTR) [h] Guaranteed value	(*)		
6.8	VFD life [years] (**)	(*)		Up to 20 years
6.10	Flanges to connect directly to an external exhaust air duct	(*)		Comply
6.11	Painting			
6.11.1	Finishing	RAL 7035		Light grey (RAL 7035)
6.11.2	Category to ISO 12944	C5-M		Comply
6.12	Dimensions (including cooling fans)			
	Length [mm]	(*)		600
	Wide [mm]	(*)		1000
	Height [mm] - panel plinth only	(*)		
	Height [mm] - total including panel plinth	(*)		2350
6.13	Weight [kg]	(*)		
6.14	Required clearances for access during operation and maintenance			
	Front [mm]	(*)		1000 mm
	Back [mm]	(*)		0 mm
	Side [mm]	(*)		Side mounting is accepted
	Top [mm]	(*)		100 mm
7.	Electrical design			
7.1.	Input:			
	Maximum short-circuit withstand current, [kA]	65 kA		Applicable with External CB or fuses

	Maximum short-circuit withstand duration [s]	1 s		Comply with External CB or fuses
	Short-circuit contribution multiplying factor for a fault at input terminal [% IN]	(*)		NA
	Rated system voltage, ULN [V]	400 V ± 10%		Comply
	Minimum voltage required for starting [% UN]	0,9		Comply
	Rated system frequency, ¦LN [Hz]	50Hz ± 5%		Comply
	Rated power, PN [kW]	200 kW		Comply
	Line-side rated a.c. current, ILN [A]	315.53 A		313 A at 400 V (normal duty)
	Fundamental line side rated a.c. current , ILN1 [A]	(*)		1.1 x In during 60 s (normal duty)
	Maximum current absorbed by VFD, considering the maximum power required in driven machine and the efficiency of all the equipment [% IN]	(*)		
	Power factor	0,95 Minimum		99 %
	Power factor at no load	(*)		(from 30 % Pn) the ATV680 leads a sinusoidal mains current with a power factor cos Phi ≈ 1
	Line-side displacement factor [cos jL1]	(*)		
	Supported neutral earthing schemes	(*)		
7.2.	Output:			
	Short-circuit contribution multiplying factor for a fault at output terminal [% IaN]	(*)		Comply
	Load-side converter a.c. rated voltage, UaN1 [V]	400 V		
	Maximum voltage variation [%]	0,1		
	Operating frequency range	(*)		0.1…500 Hz
	Maximum active power available at design temperature [kW]	(*)		Un = 400 V 217 kVA, Rated output current In 370 A , Deration factor is 2 % for each degree C
	Rated continuous output current, IaN [A]	(*)		
	Rated fundamental output current, IaN1 [A]	(*)		
	Maximum active power available at maximum temperature with derating [kW]	(*)		
	Rated continuous output current, at maximum temperature with derating IaN [A]	(*)		
	Rated fundamental output current at maximum temperature with derating, IaN1 [A]	(*)		
	Overload capability , IaM [A]	(*)		1.1 x In during 60 s (normal duty)
	Rated load power factor	(*)		Up to power factor cos Phi ≈ 1
7.3.	Number of phases	3		Comply
7.4.	Efficiency, hC [%]	97%		0.965
	hC (25%)	(*)		
	hC (50%)	(*)		
	hC (75%)	(*)		
	hC (100%)	(*)		

7.5.	Maximum guaranteed losses at rated power [kW] Guaranteed value	(*)		0.035
7.6.	Regeneration system	(*)		Not applicable on ATV680
7.7.	Speed control system	Sensorless closed loop		Comply
7.8.	Low voltage VFD: Overvoltage category according to IEC 60664	Not applicable		Category III according to EN 50178
7.9.	Earthing system	TT		6 pulse supply / TN-C or TN-S mains without neutral
7.1.	Rated insulation level			
	Switchboard and bus bars, Ui [kV]	0,6 kV		Comply
	Switching devices	0,4 kV		Comply
	Wiring:			
	Power circuits	400 V		Comply
	Control circuits	230 V		Comply
	LV AC cables color coding and labeling according IEC 60446	L1 – Brown		As per factory standard
		L2 – Black		As per factory standard
		L3 – Grey		As per factory standard
		N - Blue		As per factory standard
		PE – yellow-green		As per factory standard
	DC power supply cables color coding and labeling according IEC 60446	+		As per factory standard
		–		As per factory standard
8.	Power semiconductor			
8.1.	Rectifier			
	Type (Diode, thyristor, IGBT, etc.)	(*)		IGBT
	Total number of pulses	AFE		Comply
	Rated current	(*)		313A
	Rated voltage	(*)		400 V
	Duty class	(*)		ND
8.2.	DC Bus (type)	(*)		
8.3.	Inverter+			
	Type (Diode, thyristor, IGBT, etc.)	(*)		IGBT

	Total number of pulses	AFE		6
	Rated current	(*)		370 A
	Rated voltage	(*)		400
	Duty class	(*)		ND
9.	EMC, input and output filters			
9.1	Environmental category (IEC 61800-3)	Second		Integrated conforming to IEC 61800-3, category C3, shielded cable with 50 m Integrated conforming to IEC 61800-3, category C4, unshielded cable with 80 m
9.2	Emission category (IEC 61800-3)	C3		Comply
9.3	Input Filters	(*)		Comply
9.4	Output Filters	(*)		Comply with dv/dt filter (from 200 kW the dv/dt filter choke 150 m is built-in as standard)
9.4.1	Output dv / dt	(*)		Comply
9.4.2	Output common mode filter	(*)		NA
9.4.3	Output sinus filter	(*)		NA
9.5	Output dv / dt	(*)		Comply
9.6	Inverter commutating frequency	(*)		2.5 KHZ
9.7	All necessary filters are located within the same equipment	Yes		Yes
10.	Harmonics according to Rsc value of item 5.4			
10.1	THDi (Input)			
	At rated power	(*)		THDi is less than 2 %
	At ¾ rated power	(*)		
	At ½ rated power	(*)		
	At ¼ rated power	(*)		
10.2	THDi (Output)			
	At rated power	(*)		THDi<5 %
	At ¾ rated power	(*)		
	At ½ rated power	(*)		-
	At ¼ rated power	(*)		
10.3	Line side injected harmonics at rated power			
	3rd (%In)	(*)		Comply with IEEE519 with
	5th (%In)	(*)		
	7th (%In)	(*)		
	9th (%In)	(*)		

	9th (%In)	(*)		AFE technology
	11th (%In)	(*)		
	13th (%In)	(*)		
	15th (%In)	(*)		
	17th (%In)	(*)		
	19th (%In)	(*)		
	21st (%In)	(*)		
	23rd (%In)	(*)		
	25th (%In)	(*)		
	27th (%In)	(*)		
	29th (%In)	(*)		
	31th (%In)	(*)		
	33th (%In)	(*)		
	35th (%In)	(*)		
	37th (%In)	(*)		
	39th (%In)	(*)		
	47th (%In)	(*)		
11.	Protections			
11.1	50/51 Phase over current	Yes		Yes
11.2	27 Under voltage (L-L or L-N)	Yes		Yes
11.3	51LR Locked rotor > 100 kW	Yes		Yes
11.4	49RMS Thermal overload monitoring by PT100 inputs	Yes - Eight (8) 3-wire, Pt-100 inputs per VFD		Yes with 8 PT100 relays
11.5	66 Maximum number of starting (if it is applicable)	Yes		Yes
11.6	46 Negative-sequence overcurrent	Yes		Yes
11.7	49 Thermal relay	Yes		Yes
11.8	51N Sensitive Earth fault detection	Yes		Yes
12.	Auxiliaries(3)			
12.1	Brake resistance	(*)		NA

12.2.	Energy absorption rating of brake resistance	(*)		NA , NA , Comply , Comply with feeder for motor heater
12.3.	Re-start after voltage drop of 1 second	Yes		
12.4.	Motor heater powering and control	Yes		
12.5.	Total power consumption of all auxiliaries including cooling system [kW]	(*)		
12.6.	Tripping coil to emergency stop	Yes - EPB in front		Yes with Emergency stop push button according SIL 3 stop category 0
12.7.	Safety Integrity Level (SIL)			
	STO (Safe torque off)	Not Applicable		Comply
	SS1 (safe Stop 1)	Not Applicable		NA
	SS2 (safe Stop 2)	Not Applicable		NA
	Other safety functions	Not Applicable		NA
	Emergency stop button Category Opening breaker	No		Comply
	Safety relay with feedback	Not Applicable		NA
12.8.	Ride-through mode			
	Voltage sags down to 80% of the VFD rated voltage	≥ 5 seconds		-10 % of nominal voltage Starting the drive and continuous operation possible (1) , -15 % of nominal voltage Starting the drive and operation (1) for 10 s per 100 s possible -20 , -20 % of nominal voltage Operation (1) for less than 1 s possible , -30 % of nominal voltage Operation (1) for less than 0.5 s possible Comply with catch on fly option
	Voltage sags down to 70% of the VFD rated voltage	≥ 1 second		
	Voltage sags down to 40% of the VFD rated voltage	≥ 0,2 seconds		
	Voltage sags down to 0% of the VFD rated voltage	≥ 0,1 seconds		
12.9.	Motor flying start	Yes		
12.10.	Internal equipment			
	Input on-load switch	Yes		Comply
	High-speed fuses	Yes		Comply
	Line contactor	Yes		No need
	Heater	Yes - Hygrostatically controlled		Comply
	Lighting	Yes		Comply
	Power socket	Yes		Comply

12.11	Spare availability			
	Year of model start of manufacturing	(*)		Since 2018
	Spares availability after End of Life of the model	(*)		15 to 20 years
13.	Control and instrumentation			
13.1	Communication protocol	Modbus TCP/IP		Comply
13.2	Digital inputs/ output			
	Start / Stop input	Yes		Comply
	Start / Stop output	Yes		Comply
	Forward / Reverse output	Yes		Comply
	Acceleration / deceleration output	Yes		Comply
	Internal failure output	Yes		Comply
	External trip output	Yes		Comply
	Local / remote output	Yes		Comply
13.3	Analog inputs / outputs			
	Current output	Yes		Comply
	Motor speed output	Yes		Comply
	Speed reference input	Yes		Comply
	Voltage output	Yes		Comply
	Power output	Yes		Comply
	Torque output	Yes		Comply
	Motor frequency	Yes		Comply
	Power factor at the motor	Yes		Comply
	IGBT temperature	Yes		Comply
	Galvanic insulation for analog signals	(*)		Comply
14.	Power cables requirements			
14.1	Cables from incoming to VFD			
	Conductor	Cu		Ok
	Configuration (3-pole/single-pole)	3 pole symetrical conductors for protective grounding (delta configuration)(Note (6))		2x(3x120mm²) or 3x(3x70mm²)
	Magnetic shield	Yes		No need for Sheild
	Max cross section	(*)		
	Max number per phase	(*)		
	Insulation level	(*)		
	Maximum length (screened cable) without filters	(*)		

	Maximum length (non-screened cable) without filters	(*)		As per Cable supplier information's
	Special requirement for PE conductor	(*)		
	Special requirement when motor is in hazardous area	(*)		
	Other requisites	-		
14.2.	Cables from to motor			
	Conductor	Cu		Ok
	Configuration (3-pole/single-pole)	3 pole symetrical conductors for protective grounding (delta configuration)(Note (6))		Ok
	Magnetic shield	Yes		2x(3x240mm²) Or 4x(3x185mm²)
	Max cross section	(*)		
	Max number per phase	(*)		
	Insulation level	(*)		
	Maximum length (screened cable) without filters	(*)		
	Maximum length (non-screened cable) without filters	(*)		
	Special requirement for PE conductor	(*)		
	Special requirement when motor is in hazardous area	(*)		
	Other requisites	(*)		
14.3.	Cables from Auxiliary supply to VFD			
	Conductor	Cu		Ok
	Configuration	3-pole		
	Magnetic shield	No		
	Max cross section	(*)		
	Max number per phase	(*)		
	Insulation level	(*)		
	Special requirement for PE conductor	(*)		
	Other requisites	(*)		
14.4.	Control Cables			
	Number of wires	(*)		Comply
	Magnetic shield	Yes		
	Other requisites	(*)		
	Power consumption	(*)		
15.	Tests			
15.1.	Routine tests			
	Insulation test (IEC 60146 and 61800-5)	Yes		Ok
	Light load and functional test	Yes		Ok
	Rated current test	Yes		Ok
	Checking of auxiliary devices	Yes		Ok
	Checking the properties of the control equipment	Yes		Ok
	Checking the protective devices	Yes		Ok
15.2.	Type tests			
	Rated current	Include test certificates		Ok
	Power loss	Include test certificates		Ok
	Temperature rise test	Include test certificates		Ok
	EM immunity test	Include test certificates		Ok
	EM emission test	Include test certificates		Ok
	Current sharing	Include test certificates		Ok
	Voltage division	Include test certificates		Ok
15.3.	Special tests			
	Over current capability test	Optional - to be quoted separately		NA
	Measurement of inherent voltage regulation	Optional - to be quoted separately		NA
	Measurement of ripple voltage and current	Optional - to be quoted separately		NA
	Measurement of harmonic currents	Optional - to be quoted separately		NA

Radio frequency radiated and conducted disturbances	Optional - to be quoted separately		NA
Power factor measurement	Yes - with oscilloscope to display the VFD input wave		NA
Load tests	Yes - one (1) VFD only		Ok

	Noise test	Yes - one (1) VFD only		NA
	RideThrough test	The Bidder will provide in any case copies of the records to certificate that the VF complies with IEC 61000-4-34 or alternatively with SEMI F47		NA
15.4.	Power drive system test (special test)			
	Light load	Not Applicable		As per FAT
	Load	Not Applicable		As per FAT
	Load duty	Not Applicable		As per FAT
	Allowance FLC (full load current) vs speed	Not Applicable		As per FAT
	Temperature rise	Not Applicable		As per FAT
	Efficiency	Not Applicable		As per FAT
	Line-side current harmonic content	Not Applicable		As per FAT
	Power factor	Not Applicable		As per FAT
	Current sharing	Not Applicable		As per FAT
	Voltage division	Not Applicable		As per FAT
	Checking of auxiliary devices	Not Applicable		As per FAT
	Checking coordination of protective devices	Not Applicable		As per FAT
	Checking properties under unusual service conditions	Not Applicable		As per FAT
	Shaft current / bearing insulation	Not Applicable		As per FAT
	Audible noise	Not Applicable		As per FAT
	Motor vibration	Not Applicable		As per FAT
	EMC	Not Applicable		As per FAT
	Harmonic content of the complete drive module output current	Not Applicable		As per FAT
	Current limit and current loop	Not Applicable		As per FAT
	Speed loop	Not Applicable		As per FAT
	Torque pulsation	Not Applicable		As per FAT
	Automatic restart	Not Applicable		As per FAT
16.	Documents and drawings			
16.1	Deviations from specification(2)	(*)		Ok
16.2.	Seismic qualification test certificates/calculations	(*)		NA
16.3.	Spare parts(3)			
	For testing and commissioning	Yes		Ok
	For three years of operation (optional)	Yes		Ok
16.3.	Equipment layout with overall dimensions, weight, clearances, cable entries, fixing bolts and supports	Yes		Ok
16.4.	Efficiency losses, PF curves from no-load to 150% rated load	Yes		NA
16.5.	Electrical wiring diagrams (indicating terminal codifications)	Yes		Ok
16.6.	Operation and maintenance manuals	Yes		OK
16.7.	VFD sizing criteria according to motor and driven equipment	Yes		Ok
16.8.	Tropicalization, or any other special coating procedure	Yes		Ok
16.9.	Signal list	Yes		Ok
16.10.	Installation guide to comply with SIL category	Not Applicable		Ok
16.11.	Cabinet heat dissipation	Yes		Ok

16.12.	Detailed planning	Yes		Ok
16.13.	Quality control plan	Yes		Ok
16.14.	Detailed performance test procedures showing test schedule and techniques and including at least:			
	• Name of item to be tested.	Yes		As per FAT
	• Location of measurement points.	Yes		As per FAT
	• Manufacturer and model number of instrumentation used.	Yes		As per FAT
	• Any applicable correction curve.	Yes		As per FAT
	• Verifiable method of the test result calculation.	Yes		As per FAT
	Method to correct test conditions back to design conditions.	Yes		As per FAT
16.15.	Tests and certificates	Yes		Ok
16.16.	Final dossier	Yes		NA
16.17.	PPI Monthly report	Yes		NA

MODON ORASCOM CONSTRUCTION TROJAN COX

LV frecuency converter DSH (RO CIP - Flushing pumps)

Item	Description	Requested data	Offer	Offered data (*)
1.	General			
1.1.	Supplier	(*)		Schneider Electric
1.2.	Catalogue No.	(*)		
1.3.	KKS code	TBD		
1.4.	Manufacturer	(*)		Schneider Electric
1.5.	Place of manufacture	(*)		
1.6.	Year of manufacture	(*)		2026 or 2027
1.7.	Quantity	3		
1.8.	Applicable standards	IEC 61800-2, IEC 61800-3, IEC 61800-5-1, IEC 61800-5-2, IEC 61800-6, IEC 60146-1-1, IEC 60146-1-2, IEC 61000-4-2, IEC 61000 4-4, IEC 61000-4-5, IEC 61000-4-7, IEC 61000-4-30, IEC 62477-1, IEC 61000-2-4, IEC 61800-9-1, IEC/TR 60146		Comply with IEC 60204-1 IEC 61800-2 IEC 61800-3 IEC 61800-5-1 IEEE519
1.9.	Units of measure	SI		
2.	Scope			
2.1.	Engineering	Yes	Comply	
2.2.	Manufacturing	Yes	Comply	
2.3.	Contractual documentation to be submitted (see 7043-SPE-COX-950-734-0001_ LV Variable frequency drive TS)	Yes	Noted	
2.4.	Supply	Yes	Comply	
2.5.	Factory tests	Yes	Comply	
2.6.	Transport to site	Yes	Noted	
2.7.	Unloading	Not applicable	NA	
2.8.	Assembly	Optional	NA	
2.9.	Assembly supervision	Yes	Comply	
2.10.	Commissioning	Optional	Comply	
2.11.	Commissioning supervision	Yes	Comply	
2.12.	Spare parts for:			
	Testing and commissioning	Yes	Noted	
	Three years of operation and mantenance	Yes	Noted	
2.13.	Special tools required for operation and maintenance	Yes - if applicable	Atmospheric conditions:	
2.14.	Training program for operation and maintenance	Yes	Comply	

2.15.	Required software and any special cable	Yes	Comply
3.	Site conditions(2)		
3.1.	Site coordinates	Latitud: 31.106° N | Longitud: 27.902° E	Comply
3.2.	Location	Ras El-Hekma, Egypt	NA
3.3.	Atmospheric conditions:		
	Maximum ambient temperature [ºC]	40ºC	Comply
	Minimum ambient temperature [ºC]	0ºC	Comply
	Altitude [m]	<1000	Comply
	Relative humidity [%]	95%	Comply

	Seismic level	Moderate (Zone 2, ECP 201)	NA
	Pollution level (IEC 60664-1)	-	2 conforming to IEC 61800-5-1
3.4.	Indoor climatic conditions:		
	Maximum ambient temperature [ºC] - Air conditioned rooms	40ºC	Comply
	Maximum ambient temperature [ºC] - Not in air conditioned rooms	40ºC	Comply
	Minimum ambient temperature [ºC]	0ºC	Comply
	Relative humidity [%]	95%	Comply without condensation conforming to IEC 60068-2-3
3.5.	Design temperatures for VFD rating:		
	Indoors [ºC]	40ºC	Comply
	Maximum operation temperature (without derating)	40ºC	Comply
	Maximum operation temperature (without derating)	(*)	40 C
	Output current derating vs operation temperature factor (%/ºC)	(*)	2 % as per the derating curve
	Outdoors [ºC]	Not applicable	NA
3.6.	Classification of environmental conditions (IEC 60721-3):		
	Storage	1K2/ 1B2/ 1S1/1M1	Ambient air temperature for storage -25…70 °C
	Transportation	2K7/ 2B2/ 2S1/2M4	Transport temperature -25...+70 °C
	Stationary use at weather protected locations	3M1	Noted
	Stationary use at non-weather protected locations	Not applicable	NA
3.7.	Transformer designed and sized according to all site conditions	(*)	
4.	Hazardous area		
4.1	Hazardous area classification (IEC 60079)	Not applicable	NA
5.	Voltage supply and network impedance		
5.1	VFD Power supply	400 V ± 10%	Comply
5.2	Available auxiliaries power supplies(2) LV a.c. (non UPS)(energy consumed (VA)) UPS (only for control)(energy consumed (VA)) DC (only for control)(energy consumed (VA))	 400/230 Vac±10% 230 Vac±10% 125 Vdc+10%-15%	Comply except 125Vdc
5.3	Short circuit power at point of coupling, Scc [MVA]	(*)	Will be supplied later
5.4	Short circuit ratio, Rsc	(*)	50 kA (100 ms) with pre-fuse or circuit breaker
5.5	Overvoltage category to IEC 61800-5.	Overvoltage Category 3	Category III according to EN 50178
6	Constructive design		
6.1	Installation	Indoors	Noted

6.2	Type	cabinet-built		Comply
6.3	Protection degree of the enclosure	IP-41		IP54
6.4	Cooling system :			
	Type	Air		Air cooled with forced
	Fans (nº x %)	(*)		Air cooled with forced ventilation Power part 2320m3/h Control part 280m3/h
	Air flow [m3/h]	(*)		
	Pumps (nº x %)	(*)		
	Water consumption [l/s]	Not applicable		NA
	Water pressure acceptable at inlet	Not applicable		NA
	Heat dissipation [kW]	(*)		13060W
6.5	Sound power level, L_W [dB (A) at 1 m] Guaranteed value	(*)		73dB(A)
6.6	Mean time between failures (MTBF) [h] Guaranteed value	(*)		MTBF(Continuous Operation 66 000 h) & Intermittent Operation 89 000 h
6.7	Mean time to repair (MTTR) [h] Guaranteed value	(*)		
6.8	VFD life [years] (**)	(*)		Up to 20 years
6.10	Flanges to connect directly to an external exhaust air duct	(*)		Comply
6.11	Painting			
6.11.1	Finishing	RAL 7035		Light grey (RAL 7035)
6.11.2	Category to ISO 12944	C5-M		Comply
6.12	Dimensions (including cooling fans)			
	Length [mm]	(*)		600 mm
	Wide [mm]	(*)		1000 mm
	Height [mm] - panel plinth only	(*)		
	Height [mm] - total including panel plinth	(*)		2350 mm
6.13	Weight [kg]	(*)		
6.14	Required clearances for access during operation and maintenance			
	Front [mm]	(*)		1000 mm
	Back [mm]	(*)		0 mm
	Side [mm]	(*)		Side mounting is accepted
	Top [mm]	(*)		100 mm

7.	Electrical design			
7.1.	Input:			
	Maximum short-circuit withstand current, [kA]	65 kA		Applicable with External CB or fuses
	Maximum short-circuit withstand duration [s]	1 s		Comply with External CB or fuses
	Short-circuit contribution multiplying factor for a fault at input terminal [% IN]	(*)		NA
	Rated system voltage, ULN [V]	400 V ± 10%		Comply
	Minimum voltage required for starting [% UN]	0,9		Comply
	Rated system frequency, ¦LN [Hz]	50Hz ± 5%		Comply
	Rated power, PN [kW]	315 KW		Comply
	Line-side rated a.c. current, ILN [A]	498.53 A		491 A at 400 V (normal duty)
	Fundamental line side rated a.c. current , ILN1 [A]	(*)		1.1 x In during 60 s (normal duty)
	Maximum current absorbed by VFD, considering the maximum power required in driven machine and the efficiency of all the equipment [% IN]	(*)		
	Power factor	0,95 Minimum		99 %
	Power factor at no load	(*)		(from 30 % Pn) the ATV680 leads a sinusoidal mains current with a power factor cos Phi ≈ 1
	Line-side displacement factor [cos jL1]	(*)		
	Supported neutral earthing schemes	(*)		
7.2.	Output:			
	Short-circuit contribution multiplying factor for a fault at output terminal [% IaN]	(*)		Comply
	Load-side converter a.c. rated voltage, UaN1 [V]	400 V		
	Maximum voltage variation [%]	0,1		
	Operating frequency range	(*)		0.1…500 Hz
	Maximum active power available at design temperature [kW]	(*)		Un = 400 V 174 kVA , Rated output current In 302 A , Deration factor is 2 % for each degree C
	Rated continuous output current, IaN [A]	(*)		
	Rated fundamental output current, IaN1 [A]	(*)		
	Maximum active power available at maximum temperature with derating [kW]	(*)		
	Rated continuous output current, at maximum temperature with derating IaN [A]	(*)		
	Rated fundamental output current at maximum temperature with derating, IaN1 [A]	(*)		
	Overload capability , IaM [A]	(*)		1.1 x In during 60 s (normal duty)
	Rated load power factor	(*)		Up to power factor cos Phi ≈ 1
7.3.	Number of phases	3		Comply

7.4.	Efficiency, hC [%]	(*)		Up to 97 %
	hC (25%)	(*)		
	hC (50%)	(*)		
	hC (75%)	(*)		
	hC (100%)	(*)		
7.5.	Maximum guaranteed losses at rated power [kW] Guaranteed value	(*)		3 %
7.6.	Regeneration system	(*)		Not applicable on ATV680
7.7.	Speed control system	Sensorless closed loop		Comply
7.8.	Low voltage VFD: Overvoltage category according to IEC 60664	Not applicable		Category III according to EN 50178
7.9.	Earthing system	TT		6 pulse supply / TN-C or TN-S mains without neutral
7.1.	Rated insulation level			
	Switchboard and bus bars, Ui [kV]	0,6 kV		Comply
	Switching devices	0,4 kV		Comply
	Wiring:			
	Power circuits	400 V		Comply
	Control circuits	230 V		Comply
	LV AC cables color coding and labeling according IEC 60446	L1 – Brown		As per factory standard
		L2 – Black		As per factory standard
		L3 – Grey		As per factory standard
		N - Blue		As per factory standard
		PE – yellow-green		As per factory standard
	DC power supply cables color coding and labeling according IEC 60446	+		As per factory standard
		–		As per factory standard
8.	Power semiconductor			
8.1.	Rectifier			
	Type (Diode, thyristor, IGBT, etc.)	(*)		IGBT
	Total number of pulses	AFE		Comply
	Rated current	(*)		491A
	Rated voltage	(*)		400 V
	Duty class	(*)		ND

8.2	DC Bus (type)	(*)		DC Capacitors
8.3	Inverter+			
	Type (Diode, thyristor, IGBT, etc.)	(*)		IGBT
	Total number of pulses	AFE		6
	Rated current	(*)		590A
	Rated voltage	(*)		400
	Duty class	(*)		ND
9.	EMC, input and output filters			
9.1	Environmental category (IEC 61800-3)	Second		Integrated conforming to IEC 61800-3, category C3, shielded cable with 50 m Integrated conforming to IEC 61800-3, category C4, unshielded cable with 80 m
9.2	Emission category (IEC 61800-3)	C3		Comply
9.3	Input Filters	(*)		Comply
9.4	Output Filters	(*)		Comply with du/dt filter of 150 m cable
9.4.1	Output dv / dt	(*)		Comply
9.4.2	Output common mode filter	(*)		NA
9.4.3	Output sinus filter	(*)		NA
9.5	Output dv / dt	(*)		Comply
9.6	Inverter commutating frequency	(*)		2.5 KHZ
9.7	All necessary filters are located within the same equipment	Yes		Yes
10.	Harmonics according to Rsc value of item 5.4			
10.1	THDi (Input)			
	At rated power	(*)		THDi is less than 2 %
	At ¾ rated power	(*)		
	At ½ rated power	(*)		
	At ¼ rated power	(*)		
10.2	THDi (Output)			
	At rated power	(*)		THDi<5 %
	At ¾ rated power	(*)		
	At ½ rated power	(*)		
	At ¼ rated power	(*)		
10.3	Line side injected harmonics at rated power			
	3rd (%In)	(*)		Comply with IEEE519 with
	5th (%In)	(*)		
	7th (%In)	(*)		
	9th (%In)	(*)		

	9th (%In)	(*)		AFE technology
	11th (%In)	(*)		
	13th (%In)	(*)		
	15th (%In)	(*)		
	17th (%In)	(*)		
	19th (%In)	(*)		
	21st (%In)	(*)		
	23rd (%In)	(*)		
	25th (%In)	(*)		
	27th (%In)	(*)		
	29th (%In)	(*)		
	31th (%In)	(*)		
	33th (%In)	(*)		
	35th (%In)	(*)		
	37th (%In)	(*)		
	39th (%In)	(*)		
	47th (%In)	(*)		
11.	Protections			
11.1	50/51 Phase over current	Yes		Yes
11.2	27 Under voltage (L-L or L-N)	Yes		Yes
11.3	51LR Locked rotor > 100 kW	Yes		Yes
11.4	49RMS Thermal overload monitoring by PT100 inputs	Yes - Eight (8) 3-wire, Pt-100 inputs per VFD		Yes with 8 PT100 relays

11.5	66 Maximum number of starting (if it is applicable)	Yes		Yes
11.6	46 Negative-sequence overcurrent	Yes		Yes
11.7	49 Thermal relay	Yes		Yes
11.8	51N Sensitive Earth fault detection	Yes		Yes
12.	Auxiliaries(3)			
12.1	Brake resistance	(*)		NA , NA , Comply , Comply with feeder for motor heater
12.2	Energy absorption rating of brake resistance	(*)		
12.3	Re-start after voltage drop of 1 second	Yes		
12.4	Motor heater powering and control	Yes		
12.5	Total power consumption of all auxiliaries including cooling system [kW]	(*)		
12.6	Tripping coil to emergency stop	Yes - EPB in front		Yes with Emergency stop push button according SIL 3 stop category 0
12.7	Safety Integrity Level (SIL)			
	STO (Safe torque off)	Not Applicable		Comply
	SS1 (safe Stop 1)	Not Applicable		NA
	SS2 (safe Stop 2)	Not Applicable		NA
	Other safety functions	Not Applicable		NA
	Emergency stop button Category Opening breaker	No		Comply
	Safety relay with feedback	Not Applicable		NA
12.8	Ride-through mode			
	Voltage sags down to 80% of the VFD rated voltage	≥ 5 seconds		-10 % of nominal voltage Starting the drive and continuous operation possible (1) , -15 % of nominal voltage Starting the drive and operation (1) for 10 s per 100 s possible -20 , -20 % of nominal voltage Operation (1) for less than 1 s possible , -30 % of nominal voltage Operation (1) for less than 0.5 s possible Comply with catch on fly option
	Voltage sags down to 70% of the VFD rated voltage	≥ 1 second		
	Voltage sags down to 40% of the VFD rated voltage	≥ 0,2 seconds		
	Voltage sags down to 0% of the VFD rated voltage	≥ 0,1 seconds		
12.9	Motor flying start	Yes		
12.10	Internal equipment			
	Input on-load switch	Yes		Comply
	High-speed fuses	Yes		Comply
	Line contactor	Yes		No need
	Heater	Yes - Hygrostatically controlled		Comply
	Lighting	Yes		Comply
	Power socket	Yes		Comply

12.11	Spare availability			
	Year of model start of manufacturing	(*)		Since 2018
	Spares availability after End of Life of the model	(*)		15 to 20 years
13.	Control and instrumentation			
13.1	Communication protocol	Modbus TCP/IP		Comply
13.2	Digital inputs/ output			
	Start / Stop input	Yes		Comply
	Start / Stop output	Yes		Comply
	Forward / Reverse output	Yes		Comply
	Acceleration / deceleration output	Yes		Comply
	Internal failure output	Yes		Comply
	External trip output	Yes		Comply
	Local / remote output	Yes		Comply
13.3	Analog inputs / outputs			
	Current output	Yes		Comply
	Motor speed output	Yes		Comply
	Speed reference input	Yes		Comply
	Voltage output	Yes		Comply
	Power output	Yes		Comply
	Torque output	Yes		Comply
	Motor frequency	Yes		Comply
	Power factor at the motor	Yes		Comply
	IGBT temperature	Yes		Comply
	Galvanic insulation for analog signals	(*)		Comply
14.	Power cables requirements			
14.1	Cables from incoming to VFD			
	Conductor	Cu		Ok

	Configuration (3-pole/single-pole)	3 pole symetrical conductors for protective grounding (delta configuration)(Note (6))		2x(3x185mm²)or 3x(3x120mm²)
	Magnetic shield	Yes		No need for shield As per Cable supplier information's
	Max cross section	(*)		
	Max number per phase	(*)		
	Insulation level	(*)		
	Maximum length (screened cable) without filters	(*)		
	Maximum length (non-screened cable) without filters	(*)		
	Special requirement for PE conductor	(*)		
	Special requirement when motor is in hazardous area	(*)		
	Other requisites	-		
14.2.	Cables from VFD to motor			
	Conductor	Cu		Ok
	Configuration (3-pole/single-pole)	3 pole symetrical conductors for protective grounding (delta configuration)(Note (6))		Ok
	Magnetic shield	Yes		2x(3x240mm²) Or 4x(3x185mm²)
	Max cross section	(*)		
	Max number per phase	(*)		
	Insulation level	(*)		
	Maximum length (screened cable) without filters	(*)		
	Maximum length (non-screened cable) without filters	(*)		
	Special requirement for PE conductor	(*)		
	Special requirement when motor is in hazardous area	(*)		
	Other requisites	(*)		
14.3.	Cables from Auxiliary supply to VFD			
	Conductor	Cu		Ok
	Configuration	3-pole		
	Magnetic shield	No		
	Max cross section	(*)		
	Max number per phase	(*)		
	Insulation level	(*)		
	Special requirement for PE conductor	(*)		
	Other requisites	(*)		
14.4.	Control Cables			
	Number of wires	(*)		comply
	Magnetic shield	Yes		
	Other requisites	(*)		
	Power consumption	(*)		
15.	Tests			
15.1.	Routine tests			
	Insulation test (IEC 60146 and 61800-5)	Yes		Ok
	Light load and functional test	Yes		Ok
	Rated current test	Yes		Ok
	Checking of auxiliary devices	Yes		Ok
	Checking the properties of the control equipment	Yes		Ok
	Checking the protective devices	Yes		Ok
15.2.	Type tests			
	Rated current	Include test certificates		Ok
	Power loss	Include test certificates		Ok

Temperature rise test	Include test certificates		Ok

	EM immunity test	Include test certificates		Ok
	EM emission test	Include test certificates		Ok
	Current sharing	Include test certificates		Ok
	Voltage division	Include test certificates		Ok
15.3.	Special tests			
	Over current capability test	Optional - to be quoted separately		NA
	Measurement of inherent voltage regulation	Optional - to be quoted separately		NA
	Measurement of ripple voltage and current	Optional - to be quoted separately		NA
	Measurement of harmonic currents	Optional - to be quoted separately		NA
	Radio frequency radiated and conducted disturbances	Optional - to be quoted separately		NA
	Power factor measurement	Yes - with oscilloscope to display the VFD input wave		NA
	Load tests	Yes - one (1) VFD only		Ok
	Noise test	Yes - one (1) VFD only		NA
	RideThrough test	The Bidder will provide in any case copies of the records to certificate that the VF complies with IEC 61000-4-34 or alternatively with SEMI F47		NA
15.4.	Power drive system test (special test)			
	Light load	Not Applicable		As per FAT
	Load	Not Applicable		As per FAT
	Load duty	Not Applicable		As per FAT
	Allowance FLC (full load current) vs speed	Not Applicable		As per FAT
	Temperature rise	Not Applicable		As per FAT
	Efficiency	Not Applicable		As per FAT
	Line-side current harmonic content	Not Applicable		As per FAT
	Power factor	Not Applicable		As per FAT
	Current sharing	Not Applicable		As per FAT
	Voltage division	Not Applicable		As per FAT
	Checking of auxiliary devices	Not Applicable		As per FAT
	Checking coordination of protective devices	Not Applicable		As per FAT
	Checking properties under unusual service conditions	Not Applicable		As per FAT
	Shaft current / bearing insulation	Not Applicable		As per FAT
	Audible noise	Not Applicable		As per FAT
	Motor vibration	Not Applicable		As per FAT
	EMC	Not Applicable		As per FAT
	Harmonic content of the complete drive module output current	Not Applicable		As per FAT
	Current limit and current loop	Not Applicable		As per FAT
	Speed loop	Not Applicable		As per FAT
	Torque pulsation	Not Applicable		As per FAT
	Automatic restart	Not Applicable		As per FAT
16.	Documents and drawings			
16.1	Deviations from specification(2)	(*)		ok

16.2.	Seismic qualification test certificates/calculations	(*)		NA
16.3.	Spare parts[3]			
	For testing and commissioning	Yes		Ok
	For three years of operation (optional)	Yes		Ok
16.3.	Equipment layout with overall dimensions, weight, clearances, cable entries, fixing bolts and supports	Yes		Ok
16.4.	Efficiency losses, PF curves from no-load to 150% rated load	Yes		NA
16.5.	Electrical wiring diagrams (indicating terminal codifications)	Yes		Ok
16.6.	Operation and maintenance manuals	Yes		OK
16.7.	VFD sizing criteria according to motor and driven equipment	Yes		Ok
16.8.	Tropicalization, or any other special coating procedure	Yes		Ok
16.9.	Signal list	Yes		Ok
16.10.	Installation guide to comply with SIL category	Not Applicable		Ok
				Ok
				Ok
				Ok